

Project 1: RAG over Market, Brand & Competitive Intelligence Repositories

Domain / Business Themes	Consumer & Market Insights; AI-Powered Research Operations; Market & Competitive Intelligence; Social Media Intelligence; Synthetic Audiences (virtual respondents)
Technology Stack / Concepts	Prompt engineering, NLP, RAG, chunking, embeddings, MMR, HyDE, re-ranking, metadata filters, hybrid search, citation-grounded answer generation
Project Objective	Build a C5i-style insight assistant for research and consulting teams. The assistant should answer questions across market research reports, brand tracker summaries, product reviews, campaign reports, analyst notes, and competitive intelligence documents. It should retrieve evidence, synthesize insights, compare brands/markets/segments, cite sources, and flag confidence gaps instead of giving unsupported recommendations.
Input Data / Assets	Synthetic market research reports, brand tracker summaries, competitor profiles, campaign summaries, consumer review exports, social media snippets, category trend notes, analyst-style decks
Evaluation Metrics	Retrieval precision@K, recall@K, MRR/nDCG, answer faithfulness, citation correctness, ability to compare brands across sources, latency, user-rated insight usefulness

High-Level Work Areas

#	Work Area
1	Define representative insight questions for brand, market, category, and competitor analysis.
2	Ingest research PDFs, decks, transcripts, reviews, and benchmark reports.
3	Design chunking strategy with document, brand, market, date, channel, and study-type metadata.
4	Build baseline vector RAG and compare with keyword/BM25 retrieval.
5	Add hybrid retrieval, metadata filtering, MMR diversity, HyDE query expansion, and re-ranking.
6	Implement grounded answer templates with citations, confidence labels, and evidence tables.
7	Add evaluation set covering precision@K, faithfulness, citation accuracy, and answer usefulness.
8	Create a Streamlit or notebook demo for business users.

Progressive Demo Sequence

The sequence is designed to show limitations of a baseline approach, then progressively add retrieval, hybrid search, grounding, evaluation, and governance controls.

Step	Demo Script
1	Start with a business question: 'Why is Brand A losing share among Gen Z in urban markets?'
2	Show a simple LLM answer without documents; highlight that it sounds fluent but is not grounded in client data.
3	Add keyword search over reports; ask the same question and show that exact-term search misses synonyms such as 'youth', 'college consumers', or 'metro millennials'.
4	Add basic vector RAG; ask: 'What are the top 3 reasons for decline, with evidence?' Show improved semantic matching but also show duplicate chunks and weak source diversity.
5	Add metadata filters by market, time period, brand, and research type; ask: 'Use only 2025 India urban studies.' Highlight improved relevance but note recall risk if metadata is wrong.
6	Add hybrid BM25 + embeddings with RRF; ask a jargon-heavy query containing SKU names, campaign names, and category terms. Show why hybrid retrieval outperforms pure vector search.
7	Add MMR and re-ranking; ask: 'Compare Brand A versus Brand B across awareness, consideration, and price perception.' Show better coverage and less repetitive evidence.
8	Add HyDE/query rewriting; ask a vague business question: 'What is going wrong with the brand?' Show the system generating better retrieval queries.
9	Add answer guardrails: citations, confidence, 'insufficient evidence', and source table. Ask a question not present in the corpus and show the assistant refusing to overclaim.
10	Final limitations to highlight: stale repository, chart/table extraction errors, limited access to paid/closed research, metadata quality, and possible synthesis bias.
11	Future scope: automated ingestion pipeline, multimodal table/chart extraction, freshness monitoring, feedback-based retriever tuning, evaluation dashboard, and integration with C5i-style research workflows.

One-Page Project Snapshot

Field	Details
Project	Project 1: RAG over Market, Brand & Competitive Intelligence Repositories
Use Cases	Consumer & Market Insights; AI-Powered Research Operations; Market & Competitive Intelligence; Social Media Intelligence; Synthetic Audiences (virtual respondents)
Tech Stack / Themes	Prompt engineering, NLP, RAG, chunking, embeddings, MMR, HyDE, re-ranking, metadata filters, hybrid search, citation-grounded answer generation
Build Goal	Build a C5i-style insight assistant for research and consulting teams. The assistant should answer questions across market research reports, brand tracker summaries, product reviews, campaign reports, analyst notes, and competitive intelligence documents. It should retrieve evidence, synthesize insights, compare brands/markets/segments, cite sources, and flag confidence gaps instead of giving unsupported recommendations.
Data Needed	Synthetic market research reports, brand tracker summaries, competitor profiles, campaign summaries, consumer review exports, social media snippets, category trend notes, analyst-style decks
Success Metrics	Retrieval precision@K, recall@K, MRR/nDCG, answer faithfulness, citation correctness, ability to compare brands across sources, latency, user-rated insight usefulness